

Vol 8 Chapter 3 — Lagrangian Mechanics

Keeper (author pass — deep math/physics revision)

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Chapter 3 — Lagrangian Mechanics

Level 1 — one sentence

Lagrangian mechanics (Lagrange 1788) reformulates Newton's laws as a variational principle: a system's trajectory is the one that extremizes the action $S = \int L dt$ where $L = T - V$ is the Lagrangian (kinetic minus potential energy), giving Euler-Lagrange equations $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i}$ that are coordinate-free and naturally handle constraints.

Level 2 — graduate-physicist precision

3.1 The action principle

For a system with generalized coordinates $q_i(t)$ and Lagrangian $L(q, \dot{q}, t) = T - V$:

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

The classical trajectory between fixed endpoints is the one that makes S stationary: $\delta S = 0$.

3.2 Euler-Lagrange equations

Stationarity condition gives:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0 \quad \text{for each } i$$

These are the Euler-Lagrange equations. For Cartesian coordinates with $L = (1/2)m\dot{\vec{r}}^2 - V(\vec{r})$:

$$\frac{d}{dt}(m\dot{\vec{r}}) = -\nabla V = \vec{F}$$

— Newton's second law. Lagrangian mechanics reproduces Newton in Cartesian coordinates, but generalizes to any coordinate system (spherical, cylindrical, body-fixed, etc.) without modification.

3.3 Generalized coordinates

Lagrangian formalism works in any coordinates compatible with the system's constraints. For a pendulum: use angle θ (one DOF) instead of (x, y) with constraint. For a rigid body: use Euler angles. For a double pendulum: (θ_1, θ_2) .

Number of DOFs = configuration-space dimension after eliminating constraints.

3.4 Constraints and Lagrange multipliers

Holonomic constraints $f(q, t) = 0$ reduce DOFs. Non-holonomic constraints (like rolling without slipping) require Lagrange multipliers λ added to the action:

$$L \rightarrow L + \sum_k \lambda_k f_k(q, \dot{q}, t)$$

The multipliers turn out to be the constraint forces.

3.5 Worked example: spherical pendulum

A mass m on a string of length ℓ swinging in 3D, gravity g . Use spherical coordinates (θ, ϕ) :

$$L = \frac{1}{2} m \ell^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) - mg\ell(1 - \cos \theta)$$

Euler-Lagrange in θ :

$$m \ell^2 \ddot{\theta} - m \ell^2 \sin \theta \cos \theta \dot{\phi}^2 + mg \ell \sin \theta = 0$$

Euler-Lagrange in ϕ : since L has no explicit ϕ dependence (cyclic coordinate),

$$\frac{\partial L}{\partial \dot{\phi}} = m \ell^2 \sin^2 \theta \cdot \dot{\phi} = L_z = \text{const}$$

— angular momentum about vertical is conserved (Noether $\rightarrow$ next chapter).

3.6 D'Alembert's principle

Equivalent formulation of Lagrangian mechanics:

$$\sum_i (F_i - m_i \ddot{r}_i) \cdot \delta \vec{r}_i = 0$$

for any virtual displacement $\delta \vec{r}_i$ compatible with constraints. Avoids explicit forces of constraint.

3.7 Connection to path integral and substrate

Volume 5 Chapter 5 Section 5.3-5.4: Feynman path integral expresses quantum amplitude as $\int \mathcal{D}q e^{iS/\hbar}$. The classical-stationary path (where $\delta S = 0$) dominates by stationary phase as $\hbar \rightarrow 0$ — which is the substrate's Scale-2 limit.

Substrate-mechanism reading: Lagrangian classical mechanics is the substrate's many-tick saddle-point selection. The classical trajectory is the one the substrate's Zone 2 evolution concentrates on by destructive-interference of off-classical paths.

3.8 K-audit anchors

- Vol 5 Ch 5 Section 5.3-5.4: path integral foundation
- DCCP memory: many-tick coherent sum $\rightarrow$ classical saddle point

Level 3 — 5th-grader accessibility

Instead of writing $F = ma$, Lagrangian mechanics writes one number: $L = T - V$ (kinetic minus potential energy). Then it finds the trajectory that makes the integral $S = \int L dt$ extremal (the “action principle”). Out come the same equations of motion as Newton, but in any coordinates you like — Cartesian, polar, spherical, whatever. Big advantage: constraints (like a pendulum’s fixed length) are handled naturally. Cyclic coordinates (those that don’t appear in L) automatically give conservation laws (next chapter). In BST, the Lagrangian principle is the substrate’s classical limit of the path integral: the substrate sums over all paths quantum-mechanically; in the classical limit, the action-extremizing path dominates.

What comes next

Chapter 4 develops Hamiltonian mechanics — the canonical phase-space formalism.

Where to look this up

- Goldstein Ch 2; Landau-Lifshitz Vol 1
- BST: Vol 5 Ch 5 (path integral); DCCP memory